

SGD with Momentum

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Convex sets and convex optimisation ([Note MA-067](#)); Optimizers in deep learning ([Note DL-032](#)); Exponentially weighted moving average (EWMA) ([Note DL-033](#)).
- **Leads to:** Nesterov accelerated gradient (NAG) ([Note DL-035](#)); Adam ([Note DL-038](#)).
- **Compare with:** Stochastic gradient descent ([Note DL-020](#)); Nesterov accelerated gradient (NAG) ([Note DL-035](#)).

1. Overview

Key point: Momentum keeps an exponentially weighted average of past gradients, the velocity v , and moves by it: $v_t = \beta v_{t-1} + \eta \nabla L(w_t)$, then $w_{t+1} = w_t - v_t$. When many gradients agree, the steps grow and training speeds up; the price is overshooting the minimum.

Momentum (G-1258; Polyak 1964) is the first improved **optimizer** (G-1401). Plain **gradient descent** (G-862) forgets every gradient as soon as it has used it. Momentum remembers them: if the last few gradients all point the same way, it becomes confident and moves faster in that direction, like a ball gathering speed as it rolls downhill.

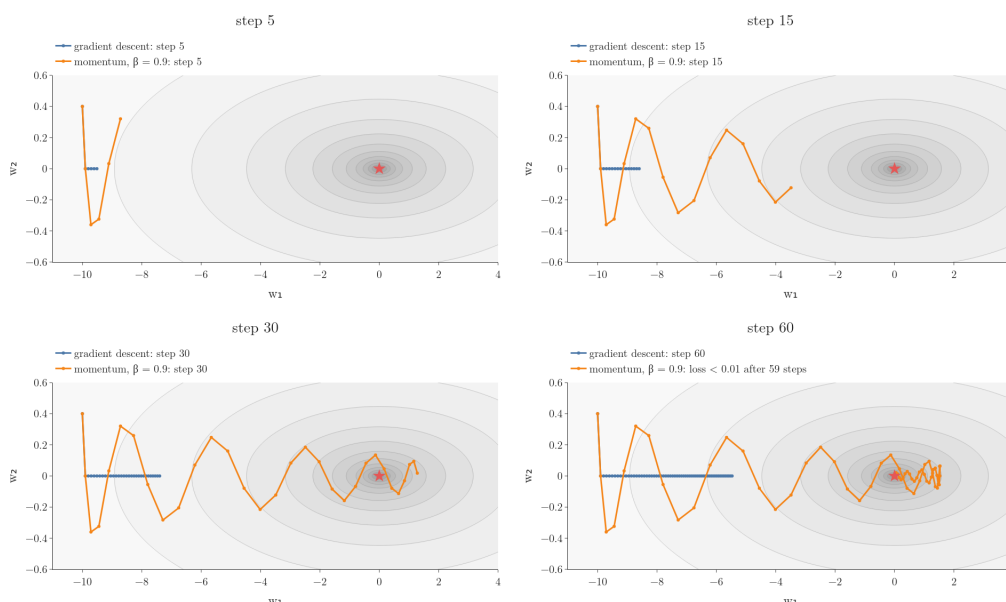


Figure 1: Plain gradient descent and momentum along a narrow valley, same learning rate 0.01. Gradient descent drops to the valley floor and crawls along it; momentum builds speed along the valley

Figure 1 shows the main benefit. On a narrow valley, gradient descent needs 424 steps to get close to the minimum; momentum needs 59. The idea of momentum returns in **NAG** (G-1315) and **Adam** (G-169), which makes it one of the most important optimizers to understand.

2. Prerequisites

- The [optimizers Note](#): the weak spots of plain gradient descent.
- The [EWMA Note](#): $V_t = \beta V_{t-1} + (1 - \beta)\theta_t$ and the role of β .
- The [gradient descent Note](#): the contour plot of a loss surface.

3. Three ways to draw a loss

Key point: One parameter: a curve. Two parameters: a surface in 3D. The same surface seen from above: a contour plot, where colour or rings show the height.

The loss is a function of the weights and biases, so we can draw it against them, just as we draw $y = f(x)$. We can only see up to 3 dimensions, so the pictures in this Note use one or two parameters:

- **One parameter, a 2D graph.** A single node with one weight w and no bias: the loss is a curve over w .
- **Two parameters, a 3D graph.** Add a bias b : the loss is a surface over the (w, b) plane, with the loss as height.
- **A contour plot** (G-468). The same surface seen from above (see section 6 of the [gradient descent Note](#)). Each ring joins points of equal loss, and colour or shading shows the height that the top view loses.

Reading a contour plot takes practice. Where the surface is flat, the height changes slowly, so the rings lie far apart. Where it is steep, they crowd together.

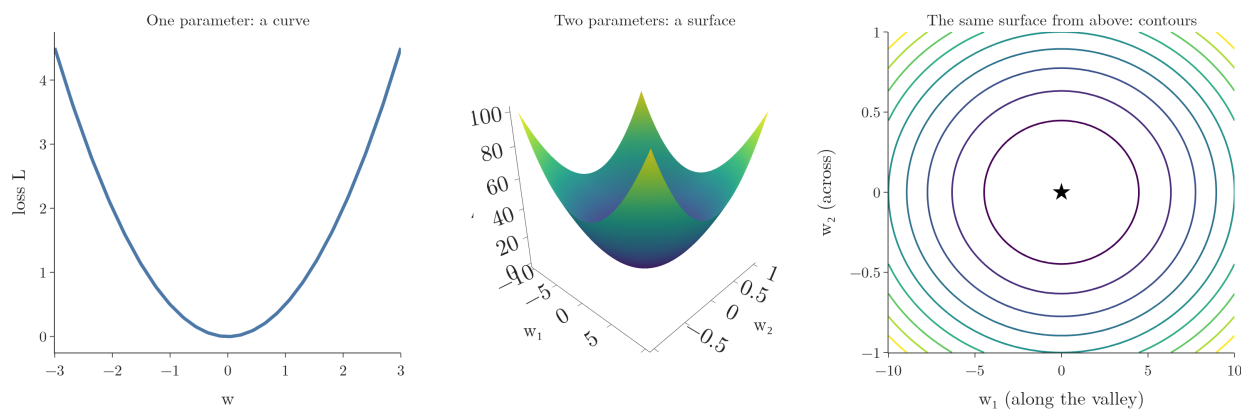


Figure 2: The three views. Left: one weight, $L(w) = w^2/2$, a curve. Middle: two weights, the narrow valley $L = (w_1^2 + 100w_2^2)/2$ of Figure 1, a surface. Right: the same valley from above; note the axes: w_2 spans only -1 to 1 .

Figure 2 draws all three. The contour plot on the right shows the valley of Figure 1: its rings look round only because the w_2 axis is stretched ten times. The loss climbs as fast over 1 unit of w_2 as over 10 units of w_1 , so across the valley the rings are packed ten times closer.

4. Why plain gradient descent struggles

Key point: Deep learning losses are non-convex. Local minima, saddle points with their flat surroundings, and high curvature all slow plain gradient descent down or trap it.

The losses of neural networks are non-convex (**non-convex functions**, G-1333; see the [convex and non-convex cost functions Note](#)), which makes the minimum hard to reach for three reasons:

1. **Local minima** (G-1110): a dip where the slope is zero. Starting from an unlucky point, gradient descent stops there and returns a sub-optimal solution.
2. **Saddle points** (G-1718): the surface rises in one direction and falls in another, and the slope changes very slowly over a wide flat region. Updates are proportional to the slope, so they become tiny there and training slows down.
3. **High curvature** (G-521): a bend with a small radius, such as the steep sides of a narrow valley. Gradient descent zigzags across the bend instead of following it.

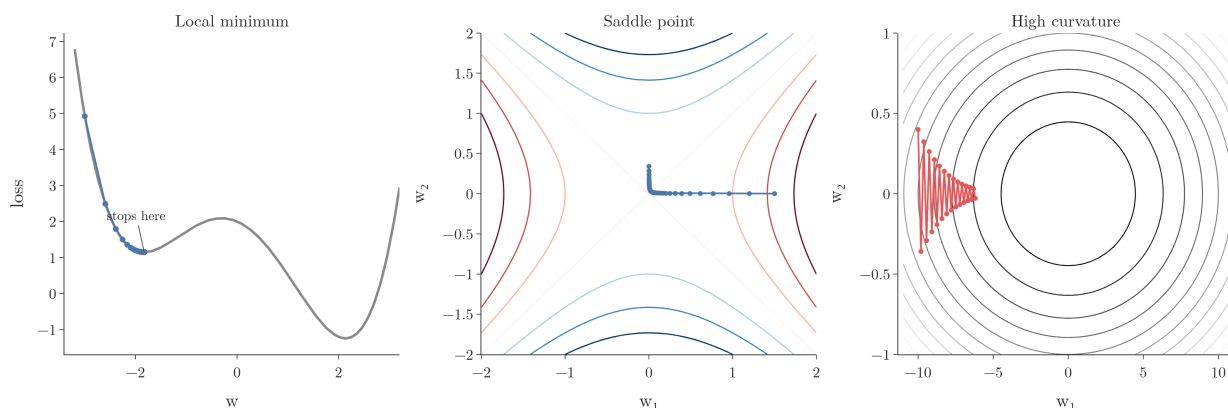


Figure 3: Plain gradient descent on the three obstacles. Left: from $w = -3$ it stops in the small dip (learning rate 0.05). Middle: on the saddle $L = w_1^2 - w_2^2$, starting almost on the ridge, its steps shrink to almost nothing near the flat centre before it slides off. Right: on the narrow valley (learning rate 0.019) it zigzags across the steep sides.

Figure 3 shows each obstacle with plain gradient descent:

- **Local minimum (left):** the curve of section 8, from $w = -3$. The steps stop at $w = -1.83$, in the small dip, although a deeper minimum lies to the right.
- **Saddle point (middle):** the first step has length 0.30; near the flat centre the steps shrink to 0.008, about 40 times shorter, before the path slowly slides off along the falling direction.

- **High curvature (right):** across the valley the slope is steep, so every step overshoots to the other side, while the progress along the valley is small.

Batch, stochastic and mini-batch gradient descent handle these poorly (see section 8 of the [gradient descent in neural networks Note](#) for their paths). Momentum was designed for exactly these situations (Goodfellow et al. 2016, §8.3.2):

- high curvature;
- small but consistent gradients;
- noisy gradients.

5. The idea: confidence builds speed

Key point: If the past gradients keep pointing the same way, move faster that way. Like a ball rolling downhill, the parameters gather speed.

An everyday picture. We drive from A to B and do not know the way, so we ask people. If four people in four places all point in the same direction, we become confident and drive faster that way. If two point forward and two point back, we still go forward, but slowly.

A physics picture. A ball rolling down a hill gathers speed as it goes. Momentum in physics is mass times velocity; with a unit mass, momentum is simply the velocity (Goodfellow et al. 2016, §8.3.2). The optimizer keeps a **velocity** (G-2085) v : the direction and speed with which the parameters move, built from the history of past updates.

The single most important benefit of momentum is speed: it usually reaches a good solution faster than plain gradient descent.

6. The update rule

Key point: The velocity is an EWMA of past gradients: $v_t = \beta v_{t-1} + \eta \nabla L(w_t)$. The weight moves by the velocity: $w_{t+1} = w_t - v_t$.

Plain gradient descent moves by the current gradient only: $w_{t+1} = w_t - \eta \nabla L(w_t)$. Momentum replaces that step by the velocity.

1. **In words:** the new velocity is a fraction β of the old velocity plus the current gradient step. The weight then moves by the whole velocity.
2. **Formula:**

$$v_t = \beta v_{t-1} + \eta \nabla L(w_t), \quad w_{t+1} = w_t - v_t$$

with $v_0 = 0$ and β between 0 and 1, usually 0.9.

3. **Example:** the loss $L(w) = w^2/2$, whose gradient is w , starting at $w_0 = -10$, with $\eta = 0.1$ and $\beta = 0.9$:

$$v_1 = 0.9 \times 0 + 0.1 \times (-10) = -1, \quad w_1 = -10 - (-1) = -9$$

$$v_2 = 0.9 \times (-1) + 0.1 \times (-9) = -1.8, \quad w_2 = -9 - (-1.8) = -7.2$$

Plain gradient descent would be at $w_2 = -9 - 0.1 \times (-9) = -8.1$. Momentum’s second step is almost twice as long, because the first step’s push is still there (Notebook).

The term βv_{t-1} is the momentum. To compute v_{t-1} we needed v_{t-2} , which needed v_{t-3} , and so on: the velocity carries the whole history of past gradients. Momentum accumulates an exponentially decaying moving average (an **EWMA**, G-735) of past gradients and continues to move in their direction (Goodfellow et al. 2016, §8.3.2).

Extra: If every gradient is the same, g , the velocity grows until it settles at a **terminal velocity** (G-1961) of $\eta g / (1 - \beta)$ (Goodfellow et al. 2016, eq. 8.17). With $\beta = 0.9$, momentum’s steps become $1 / (1 - 0.9) = 10$ times longer than plain gradient descent’s; the Notebook’s steps grow from 0.1 to 1.0. This is the $1 / (1 - \beta)$ of the [EWMA Note](#) again. Our velocity adds $\eta \nabla L$ rather than the EWMA’s $(1 - \beta) \nabla L$; that only rescales the learning rate.

Each step now has two parts: the push of the past velocity, and the current gradient. When both point the same way, the step is long.

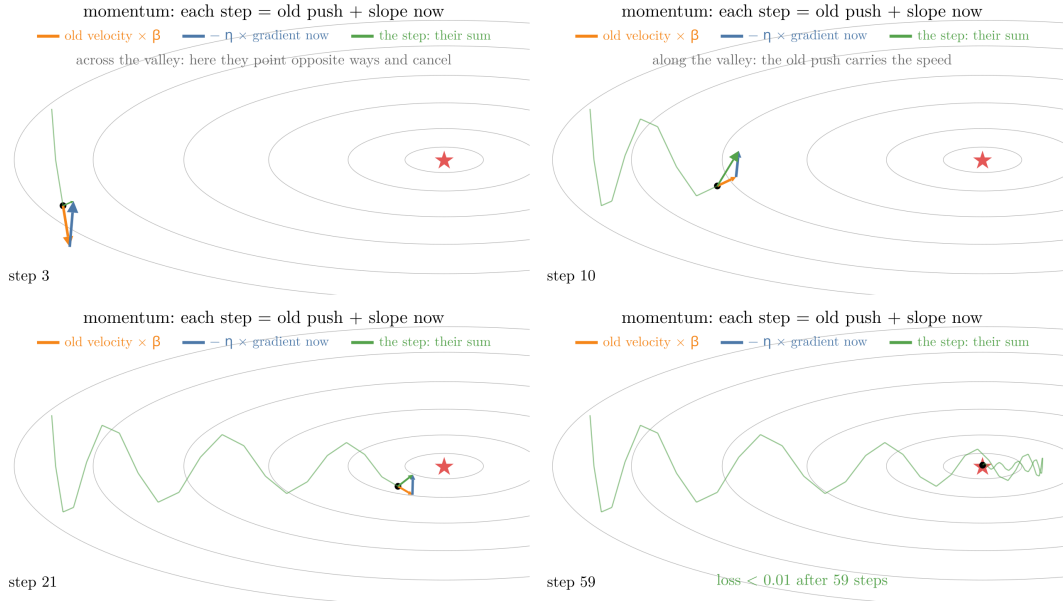


Figure 4: Momentum’s steps on the valley of Figure 1, drawn head to tail: the old velocity times β (orange), then $-\eta$ times the gradient at the current point (blue); their sum is the step (green).

Figure 4 draws each step as these two arrows placed head to tail. Watch the arrows: across the valley the orange and blue arrows often point opposite ways and cancel, while along the valley the orange arrow, the remembered push, carries most of the motion. Momentum damps the zig-zag across a narrow valley and builds speed along it (Goh 2017).

6.1 Why the zigzag fades and speed builds

Key point: Across a narrow valley the gradients keep flipping sign, so they partly cancel in the velocity. Along the valley they agree, so they add up.

In a narrow valley, plain gradient descent can bounce between the steep walls, because each step follows the current slope, which points mostly across the valley (Goodfellow et al. 2016, §8.3.2, figure 8.5). With momentum:

- **Across the valley**, the gradient changes sign as the weight swings from wall to wall. In the velocity, those opposite pushes partly cancel, so the weight stops flipping at every step.
- **Along the valley**, every gradient points the same way. Those components add up, so the velocity grows and the steps lengthen.

Whether gradient descent bounces depends on the step size. On $L = (w_1^2 + 100w_2^2)/2$ the gradient across the valley is $100w_2$, so one step gives $w_2 \leftarrow w_2 - 100\eta w_2 = (1 - 100\eta)w_2$:

- $\eta = 0.01$ (**Figure 1**): the factor is $1 - 1 = 0$. The first step puts w_2 exactly on the valley floor and it stays there; there is no zigzag, only a slow crawl along w_1 .
- $0.01 < \eta < 0.02$: the factor lies between -1 and 0 , so w_2 flips sign at every step and shrinks: a zigzag. At $\eta = 0.019$ the factor is -0.9 , and w_2 runs $0.4, -0.36, 0.324, -0.292, \dots$ (Notebook).
- $\eta > 0.02 = 2/100$: the factor is below -1 ; the zigzag grows and the run blows up.

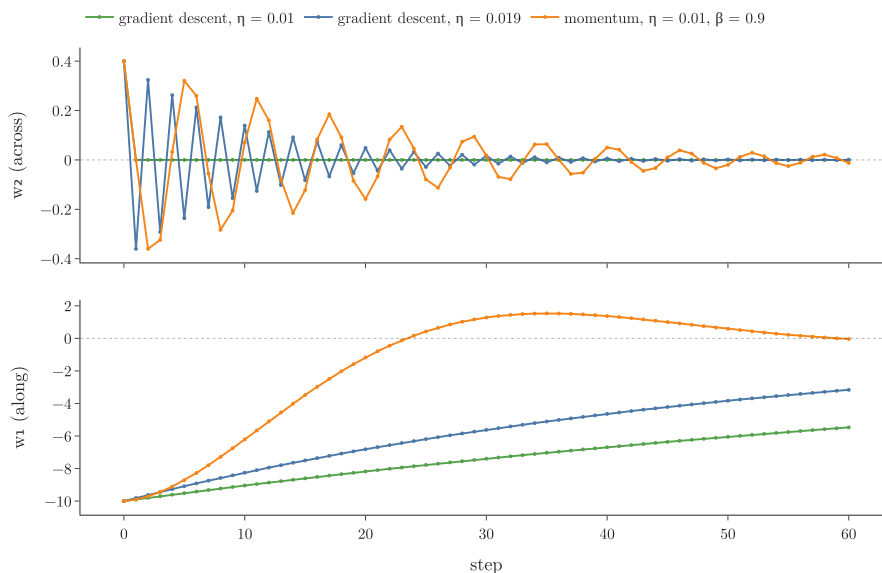


Figure 5: The first 60 steps on the valley, from $(-10, 0.4)$. Top: w_2 , the position across the valley. Bottom: w_1 , the position along it. Gradient descent at $\eta = 0.01$ (green) sits on the valley floor after one step; at $\eta = 0.019$ (blue) it zigzags. Momentum ($\eta = 0.01, \beta = 0.9$, orange) swings slowly across and races along

Figure 5 puts the three runs side by side (Notebook):

- **Gradient descent**, $\eta = 0.019$: w_2 changes sign at all 20 of the first 20 steps, moving 0.334 per step on average.
- **Momentum**, $\eta = 0.01$: w_2 changes sign 8 times and moves 0.189 per step. At step 3, for example, the remembered velocity term is +0.324 and the new gradient term is -0.36 ; they nearly cancel, so the step across is only 0.036. The swing still fades slowly: by steps 40 to 49, $|w_2|$ is still up to 0.05, against 0.006 for the zigzagging gradient descent.
- **Along the valley**, momentum covers the distance to the minimum in about 24 steps, overshoots, and comes back; both gradient descent runs are still far away after 60 steps.

The update rule explains both speeds. Write the velocity as $v_t = w_t - w_{t+1}$ and take one direction with curvature λ (1 along, 100 across), so the gradient is λw_t . Then $w_t - w_{t+1} = \beta(w_{t-1} - w_t) + \eta\lambda w_t$, that is

$$w_{t+1} = (1 + \beta - \eta\lambda) w_t - \beta w_{t-1}.$$

The two roots of $z^2 - (1 + \beta - \eta\lambda)z + \beta = 0$ multiply to β . When $(1 + \beta - \eta\lambda)^2 < 4\beta$ they are complex with equal size, so each has size $\sqrt{\beta}$: the weight swings and the swing shrinks by $\sqrt{0.9} \approx 0.95$ per step. With $\eta = 0.01$, $\beta = 0.9$ the condition holds in both directions ($0.9^2 = 0.81$ across and $1.89^2 \approx 3.57$ along, both below 3.6). Along the valley, gradient descent shrinks the distance only by $1 - 0.01 = 0.99$ per step, so momentum's 0.95 is far faster there. Across the valley, the 0.95 is slower than the 0.9 of gradient descent at $\eta = 0.019$, which is why the orange swing in Figure 5 lingers. On this valley the gain from momentum is the speed along it.

Momentum increases the step for directions whose gradients point the same way and reduces it for directions whose gradients change sign (Ruder 2016, §4.1). On the valley $L = (w_1^2 + 100w_2^2)/2$ from $(-10, 0.4)$, with $\eta = 0.01$ for both, after 20 steps gradient descent has moved w_1 from -10 to -8.18 , while momentum is already at -1.18 , close to the minimum at 0 (Notebook). To bring the loss below 0.01:

Optimizer	Steps
Gradient descent, $\eta = 0.01$	424
Gradient descent, $\eta = 0.019$ (its largest safe value)	223
Momentum, $\eta = 0.01$, $\beta = 0.9$	59

7. The role of β

Key point: $\beta = 0$ is plain gradient descent. β close to 1 remembers the past for long: more speed, more overshooting. $\beta = 1$ never forgets and never settles. Usual values: 0.5, 0.9, 0.99.

β is the **decay factor** (G-553): it decides how fast the influence of past velocities dies away. An old gradient's contribution is multiplied by β at every step, so recent gradients count most, exactly as in an EWMA. With $\beta = 0.9$ the velocity behaves roughly like an average of the last $1/(1 - 0.9) = 10$ gradients.

- $\beta = 0$: the momentum term disappears, $v_t = \eta \nabla L(w_t)$, and the update is plain gradient descent.
- $\beta = 1$: nothing decays. Like a ball on a frictionless surface, the parameters keep swinging back and forth forever without settling.
- **In practice:** 0.5, 0.9 or 0.99 (Goodfellow et al. 2016, §8.3.2).

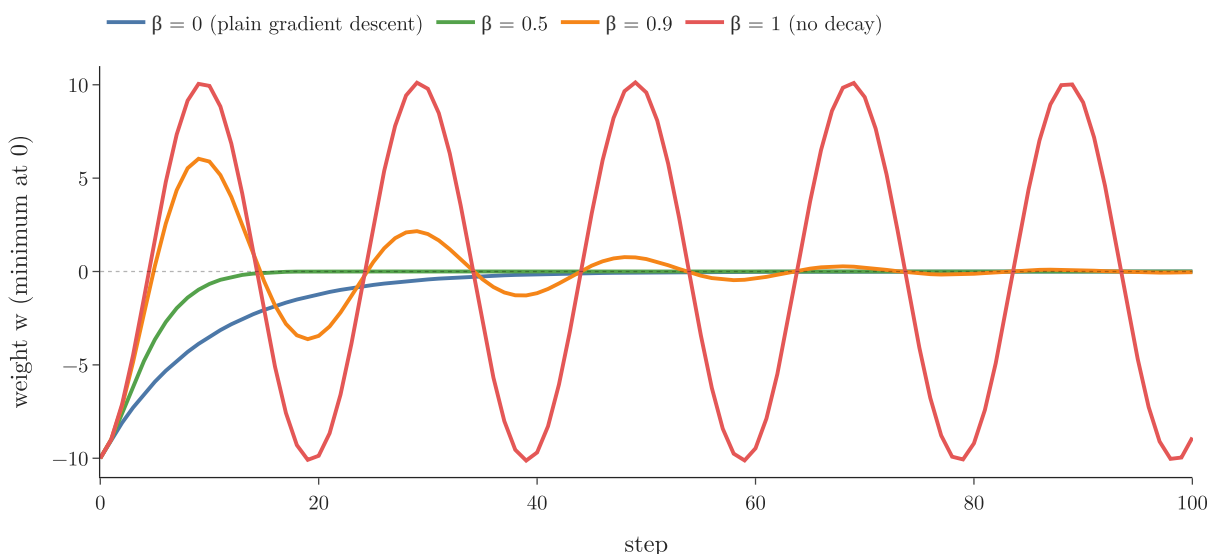


Figure 6: The weight over 100 steps on $L = w^2/2$ from $w = -10$, $\eta = 0.1$. $\beta = 0$ creeps in; $\beta = 0.5$ arrives quickly; $\beta = 0.9$ overshoots and swings before settling; $\beta = 1$ swings between -10 and 10 forever

Figure 6 shows all four. With $\beta = 1$, the weight is still swinging between about -10 and 10 after 100 steps (Notebook).

Extra: Goodfellow et al. (2016, §8.3.2) explain the decay as friction. The velocity's decay acts like viscous drag, a force proportional to $-v$, as if the particle moved through syrup. Without it ($\beta = 1$) the particle would slide down one side of the valley and up the other forever, like a hockey puck on frictionless ice.

8. Escaping a local minimum, and the cost: overshooting

Key point: The speed that carries momentum through a small dip also carries it past the true minimum. Momentum swings back and forth before settling, and those swings waste time.

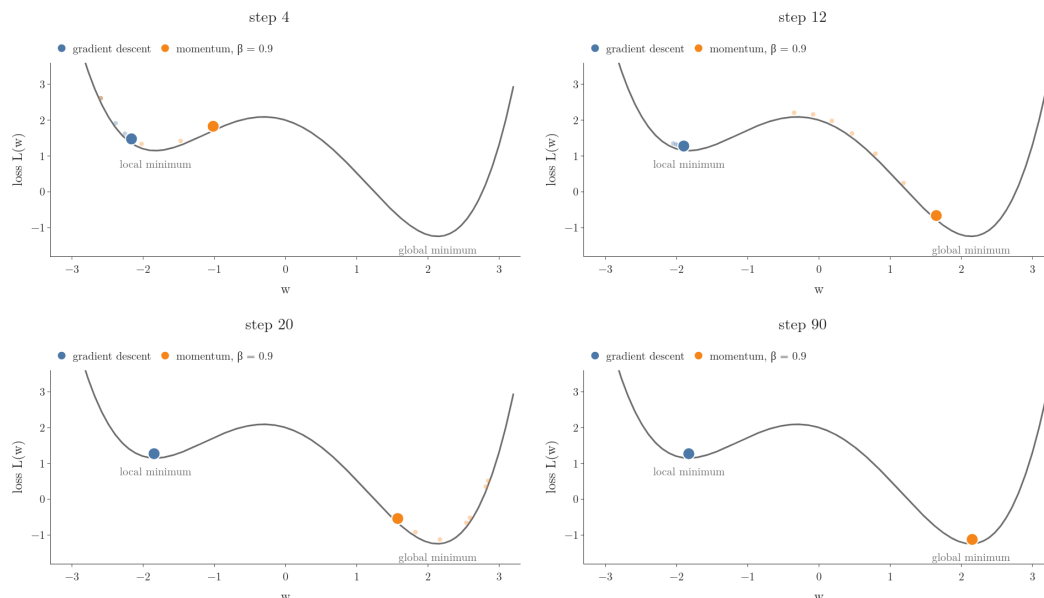


Figure 7: Two balls roll down a curve with a small dip (local minimum) and a deeper one (global minimum), $\eta = 0.05$. Plain gradient descent (blue) stops in the small dip. Momentum (orange, $\beta = 0.9$) rolls over the bump, overshoots the global minimum and swings before settling

Figure 7 shows two effects on the curve $L(w) = (w^2 - 4)^2/8 - 0.6w$, starting at $w = -3$:

1. **Faster.** The orange ball gains speed as it rolls; the blue ball moves at the pace of the slope.
2. **Out of a local minimum.** The blue ball stops in the small dip near $w = -1.83$ (loss 1.15). The orange ball has enough speed to climb out, crosses the bump, and ends in the global minimum near $w = 2.14$ (loss -1.24) (Notebook).

The same speed has a price: **overshooting** (G-1430). The orange ball does not stop at the **global minimum** (G-848): it shoots past to $w = 2.85$, comes back, and swings with shrinking amplitude before settling. In Figure 6, $\beta = 0.9$ crosses the minimum again and again for the same reason. As the decay factor makes the old velocities fade, the swings die out, but the time spent swinging is wasted.

Momentum is still faster than plain gradient descent, but these oscillations make it slower than optimizers that damp them. The biggest problem of momentum is its momentum. The next optimizer, Nesterov accelerated gradient, reduces the swings.

Extra: Escaping a dip is not guaranteed. On this curve it depends on the settings: with $\beta = 0.5$, momentum also stops in the small dip (Notebook). A ball escapes only if it has gathered enough speed before reaching the bump.

9. Momentum on real data: MNIST

Key point: On handwritten digits, adding momentum 0.9 to SGD with the same learning rate reached plain SGD's 20-epoch loss by epoch 4.

The setup:

- **data:** the **MNIST** (G-1249) handwritten digits (see the [MNIST Note](#)), 10,000 training images, each with 784 pixel **features** (G-772; input variables) and the digit as **target** (G-1949; the output we predict), and the 10,000 test images for validation;
- **network:** hidden layers of 128 and 64 **ReLU** (G-1668) nodes and a **softmax** (G-1830) output;
- **training:** mini-batch SGD (**mini-batch gradient descent**, G-1222) with **learning rate** (G-1068) 0.01, **batch size** (G-267) 64 and 20 **epochs** (G-696); only the momentum changes, 0 or 0.9.

Each is trained with 3 seeds and the curves are averaged.

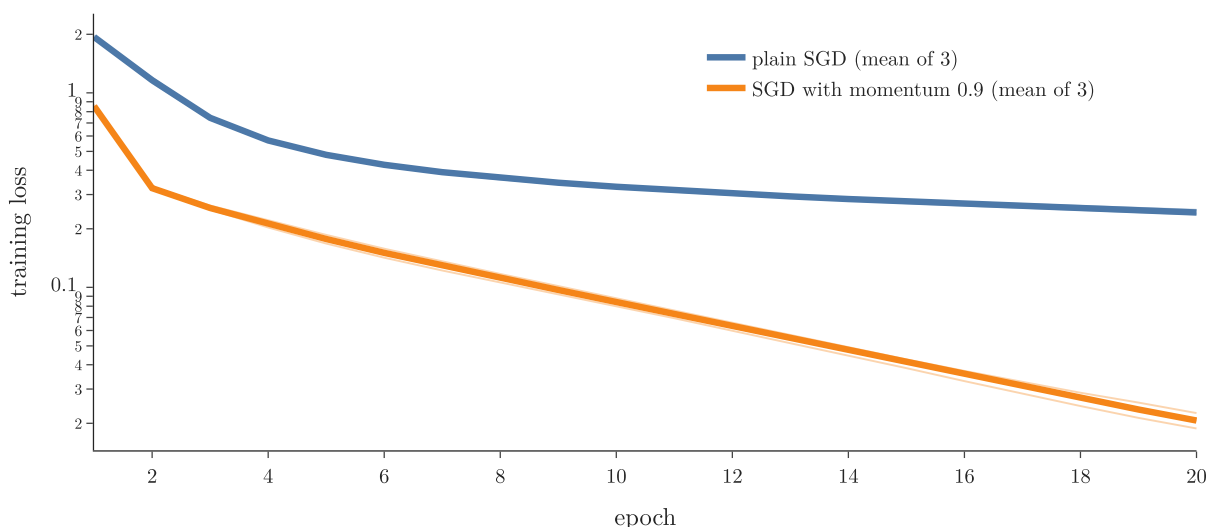


Figure 8: Training loss on MNIST per epoch, mean of 3 seeds. Same learning rate 0.01; momentum 0.9 (orange) against plain SGD (blue)

Figure 8 and the Notebook give:

	Plain SGD	SGD with momentum 0.9
Training loss, epoch 1	1.94	0.85

	Plain SGD	SGD with momentum 0.9
Training loss, epoch 5	0.48	0.18
Training loss, epoch 20	0.24	0.021
Validation accuracy, epoch 20	0.917	0.950

Momentum gets to plain SGD's final loss of 0.24 by epoch 4. Its terminal steps are up to $1/(1 - 0.9) = 10$ times longer when the gradients agree (section 6), so it covers the same ground in far fewer epochs.

9.1 Momentum in Keras

Key point: Pass momentum to Keras' SGD: 0 is plain SGD, 0.9 is SGD with momentum.

Python: SGD with momentum in Keras.

```
import keras
opt = keras.optimizers.SGD(learning_rate=0.01, momentum=0.9)
model.compile(optimizer=opt, loss="sparse_categorical_crossentropy",
               metrics=["accuracy"])
```

momentum is β , and its default is 0. With momentum=0.9 and nesterov=False (the default) Keras uses the update of section 6.

Extra: Keras writes the same update with the sign of v flipped: $v \leftarrow \beta v - \eta g$, then $w \leftarrow w + v$ (Keras SGD documentation). Both give the same weights.

10. Summary

	Plain gradient descent	SGD with momentum
Step	$\eta \nabla L(w_t)$	$v_t = \beta v_{t-1} + \eta \nabla L(w_t)$
Memory of past gradients	none	EWMA, about $1/(1 - \beta)$ steps
Narrow valley (our example)	424 steps	59 steps
Small local dip	stops in it	can roll through
Near the minimum	arrives directly	overshoots and swings

- Momentum adds a velocity, an exponentially decaying average of past gradients, to gradient descent.
- Gradients that agree add up (speed); gradients that flip sign cancel (less zigzag).
- $\beta = 0$ is plain gradient descent; $\beta = 1$ never settles; 0.9 is the usual value.
- The speed can carry it out of a small local minimum, and also past the global minimum: overshooting is its main weakness.
- In Keras: `SGD(learning_rate=..., momentum=0.9)`.

11. Sources

Built from

- CampusX, “SGD with Momentum Explained in Detail with Animations | Optimizers in Deep Learning Part 2”, YouTube, <https://www.youtube.com/watch?v=vVS4csXRlcQ>

Other references

- Goh, G. (2017). Why momentum really works. *Distill*, distill.pub/2017/momentum. Introduction: momentum’s added inertia damps oscillations and carries the descent through narrow valleys (the statement cited with Figure 4). The two-arrow drawing, data and code are our own.
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. §8.3.2 Momentum (algorithm 8.2, eq. 8.17, the physical analogy).
- Polyak, B. T. (1964). Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics* 4(5), 1–17.
- Ruder, S. (2016). An overview of gradient descent optimization algorithms. [arXiv:1609.04747](https://arxiv.org/abs/1609.04747). §4.1 Momentum.
- Keras API documentation: `SGD` optimizer, keras.io/api/optimizers/sgd.

12. Key terms

Term	Meaning
Momentum (optimizer)	Gradient descent that moves by a velocity, an exponentially decaying average of past gradients
Velocity v	The direction and size of the current move, built from past gradients: $v_t = \beta v_{t-1} + \eta \nabla L(w_t)$
Decay factor β	How much of the old velocity is kept each step; 0 gives plain gradient descent, usually 0.9
Terminal velocity	The step size momentum reaches when every gradient is the same: $\eta g / (1 - \beta)$
Overshooting	Moving past the minimum because of the built-up velocity, then swinging back
Contour plot	A loss surface seen from above, with rings joining points of equal loss
Feature	An input variable, such as one pixel of an image
Target	The output we predict, such as the digit